Week 3 build card

Attach a cone pusher and test the dance

Team ____ Builders A ____ B ____ C ____

Build a pusher that attaches to the robot and guides a small upright cone during the dance's forward moves AND turns. The cone slides on the floor; do not fasten it to the pusher or carry it.

Materials: Beams, pins, braces, short side guides or pre-cut card, matching mounting proxy and small cone. No extra motor or sensor.

Build and prepare

1. Match the two front mounting points marked by the mentor. Use the proxy while the other team has the robot.

2. Make a low pushing bar with short side guides. Try a shallow V or open U shape with room for the cone to slide. Brace the mount against twisting.

3. Check wheel and floor clearance. Hand-test a gentle forward push AND turn. Change a part that bends, tips the cone or lets it escape.

Try it on the robot

4. Attach YOUR pusher to the actual stopped robot. With the mentor, check both mounts, wheels, cables and buttons. A hand test alone does not finish this challenge.

5. Put the cone in the guide and run the SAME dance loop: drive(90), turn(90), repeated three times, then signal D once. Use await and the supplied full project. Keep the pusher attached during the turns.

6. Count complete drive-and-turn repeats before a tip, escape or loose mount (0-3). Stop for a failure; reset only after motion stops. Improve ONE feature and rerun the same code.

Success: Three complete repeats with the mount secure and the cone still upright inside the guide. Keep cone, floor, start, speed and code the same when comparing builds.

Our jobs and evidence

Rotate builder, designer and tester. At the robot, rotate driver, navigator and observer when called.

First test: ____ / 3 repeats. After one change: ____ / 3 repeats.

We noticed _____. We changed _____.

SPEED stays 200. Mentors may rehearse a smaller positive turn if needed, but turns remain in the dance. Finish testing by minute 50.